Solve with us 1.1 - Not Graded

This assignment will not be graded and is only for practice.

Key Points:

- Vectors in R^2 R2 are represented by ordered pairs (a, b)(a, b).
- The first element in the ordered pair denotes the XX-coordinate and the second one denotes the YY -coordinate in the Cartesian plane.
- The addition of two vectors $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2) \in R^2$ v1 = (x1, y1) and v2 = (x2, y2)) ∈ R2, is given by $v_1 + v_2 = (x_1 + x_2, y_1 + y_2)$v1 + v2 = (x1 + x2, y1 + y2).
- The scalar multiplication of a vector $v = (x, y) \in R^2$ v = (x, y) ∈ R2 with a scalar $c \in R$c ∈ R, is given by $cv = (cx, cy)$cv = (cx, cy).

1 point

Let $v_1 = (2, -4)$v1 = (2, −4) and $v_2 = (-1, 2)$v2 = (−1, 2) be two vectors in R^2 R2. Which of the following options are true?

[Hint: Recall that vector addition and scalar multiplication are done coordinatewise.]

- ☐ $2v_2 = (-2, 4)$2v2 = (−2, 4)
- ☐ $\frac{1}{2}v_1 = (1, -2)$21v1 = (1, −2)
- ☐ $v_1 + 2v_2 = (0, 0)$v1 + 2v2 = (0, 0)
- ☐ $2v_1 + v_2 = (0, 0)$2v1 + v2 = (0, 0)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$2v_2 = (-2, 4)$2v2 = (−2, 4)

$\frac{1}{2}v_1 = (1, -2)$21v1 = (1, −2)

$v_1 + 2v_2 = (0, 0)$v1 + 2v2 = (0, 0)

1 point

Choose the set of correct options using Figure M2W1SWU1.

[Hint: Recall that vector addition and scalar multiplication are done coordinate-wise.]

- ☐ $2A$2A is the vector (2, 42, 4).
- ☐ $3B$3B is the vector (6, 96, 9).
- ☐ $A + B$A + B is the vector (3, 53, 5).
- ☐ $A - B$A − B is the vector (−1, −1−1, −1).

No, the answer is incorrect.

Score: 0

Accepted Answers:

$2A$2A is the vector (2, 42, 4).

$3B$3B is the vector (6, 96, 9).

$A + B$A + B is the vector (3, 53, 5).

$A - B$A − B is the vector $(-1, -1-1, -1)$.

2. Key points:

- The addition of two vectors $V_1 = (a_1, b_1, C_1)$V1 = (a1, b1, C1) and $V_2 = (a_2, b_2, C_2)$V2 = (a2, b2, C2) are done coordinate-wise, as follows:
- $V_1 + V_2 = (a_1, b_1, C_1) + (a_2, b_2, C_2) = (a_1 + a_2, b_1 + b_2, C_1 + C_2)$V1 + V2 = (a1, b1, C_1) + (a2, b2, C2) = (a1 + a2, b_1 + b2, C1 + C2)

- The scalar multiplication of a vector $V = (a, b, c)$V = (a, b, c) by a scalar αα is given by:
$\alpha(a, b, c) = (\alpha a, \alpha b, \alpha c)$α(a, b, c) = (αa, αb, αc)

1 point

Let $V_1 = (1, 0, 0)$V1 = (1, 0, 0), $V_2 = (0, 1, 0)$V2 = (0, 1, 0) and $V_3 = (0, 0, 1)$V3 = (0, 0, 1) be three vectors and $a, b,$ and ca, b, and c be three real numbers (scalars). Then, which of the following is (are) true?

☐ $(a, b, c) = aV_1 + bV_2 + cV_3$(a, b, c) = aV1 + bV2 + cV3

☐ $(a, b, c) = abV_1 + bcV_2 + caV_3$(a, b, c) = abV1 + bcV2 + caV3

☐ $(a, 0, c) = aV_1 + cV_2 + 0V_3$(a, 0, c) = aV1 + cV2 + 0V3

☐ $(a, 0, c) = aV_1 + 0V_2 + cV_3$(a, 0, c) = aV1 + 0V2 + cV3

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(a, b, c) = aV_1 + bV_2 + cV_3$(a, b, c) = aV1 + bV2 + cV3
$(a, 0, c) = aV_1 + 0V_2 + cV_3$(a, 0, c) = aV1 + 0V2 + cV3

3. Key Points:

- Addition of matrices $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$A = [a11 a12 a13 a21 a22 a23 a31 a32 a33] and $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$B = [b11 b12 b13 b21 b22 b23 b31 b32 b33] is given by

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \\ a_{31} + b_{31} & a_{32} + b_{32} & a_{33} + b_{33} \end{bmatrix}$$A + B = [a11 + b11 a12 + b12 a13 + b13 a21 + b21 a22 + b22 a23 + b23 a31 + b31 a32 + b32 a33 + b33]

a22 + b22a32 + b32a13 + b13a23 + b23a33 + b33]

- If $R = [abc]$R = [abc] and $C = \begin{bmatrix} d \\ e \\ f \end{bmatrix}$C = [def], then the product RCRC is given by

  $[ad + be + cf]$[ad + be + cf]

- Suppose $A = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$A = [R1R2] and $B = [C_1\ C_2\ C_3]$B = [C1C2C3], where R_i Ri's denote the rows of matrix AA and C_i Ci's denote the columns of matrix BB. Moreover, assume that the number of columns of AA and the number of rows of BB are the same. The product ABAB is given by,

  $AB = \begin{bmatrix} R_1C_1 & R_1C_2 & R_1C_3 \\ R_2C_1 & R_2C_2 & R_2C_3 \end{bmatrix}$AB = [R1C1R2C1R1C2R2C2R1C3R2C3]

- Addition of two matrices AA and BB is defined if both AA and BB have the same number of rows and the same number of columns. If both the matrices AA and BB have mm rows and nn columns, then the matrix $A + B$A + B also has mm rows and nn columns

- If AA is an $m \times n$m × n matrix and BB is an $n \times p$n × p matrix, then ABAB is well-defined and is an $m \times p$m × p matrix.

1 point

Suppose $P = \begin{bmatrix} 3 & -1 & 7 \\ 4 & 0 & 1 \\ 2 & -5 & 2 \end{bmatrix}$P = [342−10−5712], $Q = [14 -9]$Q = [14−9], $R = [0 -3 10]$R = [0−310], $D = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$D = [−245]

[Hint: If AA is a matrix of order $m \times n$m × n and BB is a matrix of order $n \times p$n × p, then the order of ABAB is $m \times p$m × p.]

- ☐ The matrix PDPD is of order $3 \times 1$3 × 1.
- ☐ The matrix PDPD is of order $1 \times 3$1 × 3.
- ☐ The matrix QDQD is of order $3 \times 3$3 × 3.
- ☐ The matrix QDQD is of order $1 \times 1$1 × 1.
- ☐ The matrix DQDQ is of order $3 \times 3$3 × 3.
- ☐ The matrix DQDQ is of order $1 \times 1$1 × 1.
- ☐ The product QDQD is not defined.
- ☐ The product QRQR is not defined.

The addition $P + Q$P + Q is not defined.

☐

The addition $P + D$P + D is not defined.

No, the answer is incorrect.

Score: 0

Feedback:

• P is a 3 × 3 matrix, D is a 3 × 1 matrix. Think about the order of P D.
• Q is a 1 × 3 matrix, D is a 3 × 1 matrix. Think about the order of QD and DQ.
• Q is a 1 × 3 matrix, R is a 1 × 3 matrix. Think whether is it possible to define QR or not.
• Figure out which pair of matrices has the same number of rows as well as the same number of columns.

Accepted Answers:

The matrix PDPD is of order $3 \times 1$3 × 1.
The matrix QDQD is of order $1 \times 1$1 × 1.
The matrix DQDQ is of order $3 \times 3$3 × 3.
The product QRQR is not defined.
The addition $P + Q$P + Q is not defined.
The addition $P + D$P + D is not defined.

1 point

Suppose $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 5 & 3 \\ 3 & 3 & 4 \end{bmatrix}$A = $\begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 2 & 3 & 3 \\ 2 & 5 & 4 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 2 & 3 \\ 2 & 5 & 4 \end{bmatrix}$B = $\begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 2 & 5 & 3 \\ 2 & 5 & 3 & 4 \end{bmatrix}$. Which of the following options are correct?

[Hint: Multiplication of matrices $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$A = $\begin{bmatrix} a_{11} & a_{21} & a_{31} & a_{12} & a_{22} & a_{32} & a_{13} & a_{23} & a_{33} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$B = $\begin{bmatrix} b_{11} & b_{21} & b_{31} & b_{12} & b_{22} & b_{32} & b_{13} & b_{23} & b_{33} \end{bmatrix}$ is given by

$AB = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} & a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} & a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} \\ a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} & a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} & a_{31}b_{13} + a_{32}b_{23} + a_{33}b_{33} \end{bmatrix}$AB = $\begin{bmatrix} a11b11 + a12b21 + a13b31 & a21b11 + a22b21 + a23b31 & a31b11 + a32b21 + a33b31 & a11b12 + a12b22 \\ + a13b32 & a21b12 + a22b22 + a23b32 & a31b12 + a32b22 + a33b32 & a11b13 + a12b23 + a13b33 & a21b13 \\ + a22b23 + a23b33 & a31b13 + a32b23 + a33b33 \end{bmatrix}$ Similarly, BABA can be calculated.]

☐

$AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 26 & 29 \\ 11 & 29 & 34 \end{bmatrix}$AB = $\begin{bmatrix} 5 & 10 & 11 & 10 & 26 & 29 & 11 & 29 & 34 \end{bmatrix}$

☐

$AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 29 & 26 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 6 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 51 & 111 \\ 102 & 926 \end{bmatrix}$ AB = [5 10 11 / 10 29 26] [1 1 / 2 6 / 1 2] = [51 111 / 102 926]

☐

$BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 13 & 22 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 4 & 2 \\ 2 & 45 \end{bmatrix} = \begin{bmatrix} 109 & 1491 \\ 142 & 245 \end{bmatrix}$ BA = [10 9 14 / 9 13 22] [1 1 / 4 2 / 2 45] = [109 1491 / 142 245]

☐

$BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 22 & 13 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 4 & 2 \\ 1 & 45 \end{bmatrix} = \begin{bmatrix} 109 & 1492 \\ 141 & 345 \end{bmatrix}$ BA = [10 9 14 / 9 22 13] [1 1 / 4 2 / 1 45] = [109 1492 / 141 345]

No, the answer is incorrect.
Score: 0
Feedback:
Observe that B = A^T (where A^T denotes the transpose of A).
Accepted Answers:
$AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 29 & 26 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 6 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 51 & 111 \\ 102 & 926 \end{bmatrix}$ AB = [5 10 11 / 10 29 26] [1 1 / 2 6 / 1 2] = [51 111 / 102 926]

$BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 13 & 22 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 4 & 2 \\ 2 & 45 \end{bmatrix} = \begin{bmatrix} 109 & 1491 \\ 142 & 245 \end{bmatrix}$ BA = [10 9 14 / 9 13 22] [1 1 / 4 2 / 2 45] = [109 1491 / 142 245]

1 point

Suppose $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ A = [1 0 / 0 −1] and $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ B = [0 1 / 0 0] Which of the following options are true?

[Hint: Calculate A^2 A2 and B^2 B2]

☐

$A^2 = I$ A2 = I

☐

$A^2 = A$ A2 = A

☐

$B^2 = I$ B2 = I

☐

$B^2 = 0$ B2 = 0

No, the answer is incorrect.
Score: 0
Feedback:
Correctly solving this question will give a hint to solve Week 4 Lecture 1.2 : Activity Question 6.
Accepted Answers:
$A^2 = I$ A2 = I

$B^2 = 0$ B2 = 0

4. Key points:

- Consider a system of linear equations as follows:
$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$ a11x1 + a12x2 + a13x3 = b1a21x1 + a22x2 + a23x3
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

= b2a31x1 + a32x2 + a33x3 = b3

Let the matrix representation of the above system be $Ax = b$Ax = b, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$A = [a11 a21a31a12a22a32a13a23a33], $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = [x1x2x3], and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$b = [b1b2b3].

- AA is called Coefficient matrixCoefficient matrix.

- Suppose there are mm number of equations and nn number of variables in the system of linear equations, then the coefficient matrix will be an $m \times n$m × n matrix, xx will be an $n \times 1$n × 1 and bb will be an $m \times 1$m × 1 matrix.
1 point
Consider a system of linear equations (System 1):

$$\begin{aligned} -2x_1 + 3x_2 + x_3 &= 1 \\ -x_1 + x_3 &= 0 \\ 2x_2 &= 5 \end{aligned}$$ −2x1 + 3x2 + x3−x1 + x32x2= 1= 0= 5

If the matrix representation of system (1) is $Ax = b$Ax = b, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = [x1x2x3], then

[Hint: If a variable is absent in an equation then the coefficient of the variable must be taken as 0.]

☐ $A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$A = [−231−101020]

☐ $b = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$b = [501]

☐ $A = \begin{bmatrix} -2 & -1 & 0 \\ 3 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix}$A = [−2−103021110], and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$b = [105]

☐

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$A = [−231−101020], and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$b = [105]

No, the answer is incorrect.
Score: 0
Accepted Answers:

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$A = [−231−101020]

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$A = [−231−101020], and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$b = [105]

1 point
Consider a system of equations:

$$\begin{aligned} 2x_1 + 3x_2 &= 6 \\ -2x_1 + kx_2 &= d \\ 4x_1 + 6x_2 &= 12 \end{aligned}$$ 2x1 + 3x2−2x1 + kx24x1 + 6x2 = 6 = d = 12

Choose the set of correct options.

[Hint: Observe that third equation is a multiple of the first one (Dividing by 2 from both the sides of the third equation gives the first equation). So it is enough to check the solutions for the first and second equation.]

☐ $Ax = b$Ax = b represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$x = [x1x2], $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$A = [2−243k66], and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$b = [6d12]

☐ The system has no solution if $k = -3$k = −3, $d = 0$d = 0.

☐ The system has a unique solution if $k = 3$k = 3, $d = 0$d = 0.

☐ The system has infinitely many solutions if $k = -3$k = −3, $d = 6$d = 6.

☐ The system has infinitely many solutions if $k = -3$k = −3, $d = -6$d = −6.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$Ax = b$Ax = b represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$x = [x1x2], $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$A = [2−243k6], and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$b = [6d12]

The system has no solution if $k = -3$k = −3, $d = 0$d = 0.
The system has a unique solution if $k = 3$k = 3, $d = 0$d = 0.
The system has infinitely many solutions if $k = -3$k = −3, $d = -6$d = −6.

5. Key points:

- Let AA be a $2 \times 2$2 × 2 matrix as follows:

  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$[acbd]

  $det(A) = ad - bc$det(A) = ad − bc

- Let AA be a $3 \times 3$3 × 3 matrix as follows:

  $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$[a11a21a31a12a22a32a13a23a33]

  $det(A) = a_{11} \times det(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}) - a_{12} \times det(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}) + a_{13} \times det(\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix})$det(A) = a11 × det ([a22a32a23a33]) − a12 × det ([a21a31a23a33]) + a13 × det ([a21a31a22a32])

  i.e., $det(A) = a_{11} \times (a_{22}\,a_{33} - a_{32}\,a_{23}) - a_{12} \times (a_{21}\,a_{33} - a_{23}\,a_{31}) + a_{13} \times (a_{21}\,a_{32} - a_{22}\,a_{31})$ det(A) = a11 × (a22a33 − a32a23) − a12 × (a21a33 − a23a31) + a13 × (a21a32 − a22a31)
  (Expanding with respect to first row)
  Find the determinant of the matrix $\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$[3121]

  [Hint: Use the formula for finding the determinant of a matrix of order $2 \times 2$2 × 2.]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
1 point

Let $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix}$A = [322232211] be a $3 \times 3$3 × 3 matrix. Which of the following is(are) correct? [Hint: Determinant can be calculated by expanding with respect to different rows or columns.]

- $det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$det(A) = 3 × det ([3121]) − 2 × det ([2221

]) + 2 × det ([2231])

☐ $det(A) = 3 \times det([\begin{smallmatrix}3&2\\1&1\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2&2\\1&2\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2&3\\2&1\end{smallmatrix}])$det(A) = 3 × det ([3121]) + 2 × det ([2122

]) + 2 × det ([2231])

☐ $det(A) = -2 \times det([\begin{smallmatrix}2&2\\1&1\end{smallmatrix}]) + 3 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}]) - 2 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}])$det(A) = −2 × det ([2121]) + 3 × det ([3221

]) − 2 × det ([3221])

☐ $det(A) = 2 \times det([\begin{smallmatrix}2&2\\1&1\end{smallmatrix}]) - 3 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}])$det(A) = 2 × det ([2121]) − 3 × det ([3221

]) + 2 × det ([3221])

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A) = 3 \times det([\begin{smallmatrix}3&2\\1&1\end{smallmatrix}]) - 2 \times det([\begin{smallmatrix}2&2\\2&1\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2&3\\2&1\end{smallmatrix}])$det(A) = 3 × det ([3121]) − 2 × det ([2221

]) + 2 × det ([2231])

$det(A) = 3 \times det([\begin{smallmatrix}3&2\\1&1\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2&2\\1&2\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2&3\\2&1\end{smallmatrix}])$det(A) = 3 × det ([3121]) + 2 × det ([2122

]) + 2 × det ([2231])

$det(A) = -2 \times det([\begin{smallmatrix}2&2\\1&1\end{smallmatrix}]) + 3 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}]) - 2 \times det([\begin{smallmatrix}3&2\\2&1\end{smallmatrix}])$det(A) = −2 × det ([2121]) + 3 × det ([3221

]) − 2 × det ([3221])

6. Key points:

- Row operations and relation with determinants:

  - Type 1: Interchanging two rows of a matrix changes the sign of the determinant.
  - Type 2: Multiplying a real number with a row and adding it to some other row does not change the determinant.
  - Type 3: The determinant of a new square matrix obtained by multiplying a real number cc with a row of square matrix of order nn, is cc times the determinant of the original matrix.

- Same types of column operations, as discussed above for row operations, gives same type of relation with the determinants.
- If two rows or two columns of a matrix are equal, then the determinant of the matrix is 00.
- If all the elements of any row or any column of a matrix are 00, then the determinant of the matrix is 0 0.
- $det(A) = det(A^T)$det(A) = det(AT), where A^T AT denotes the transpose of AA.
- $det(AB) = det(BA) = det(A)det(B)$det(AB) = det(BA) = det(A)det(B).

1 point
Let AA be a $3 \times 3$3 × 3 matrix with non-zero determinant. If $det(2A) = k\ det(A)$det(2A) = k det(A), then what will be the value of kk?
[Hint: If a scalar (c)(c) is multiplied with one row of a matrix AA, then the determinant of the new matrix will be cc times the determinant of AA.]

○
22
○
44
○
88
○
1616

No, the answer is incorrect.
Score: 0
Accepted Answers:
88

1 point

Let AA be a $2 \times 22 \times 2$ matrix, which is given as $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$[a11a21a12a22]. Define the following matrices :

$B = \begin{bmatrix} a_{11} & -a_{21} \\ a_{12} & -a_{22} \end{bmatrix}$B = [a11 − a21a21a12 − a22a22], $C = \begin{bmatrix} a_{11} & -a_{12} \\ a_{21} & -a_{22} \end{bmatrix}$C = [a11 − a12a21 − a22a12a22],

$D = \begin{bmatrix} a_{11} & +a_{21} \\ a_{12} & -a_{22} \end{bmatrix}$D = [a11 + a21a21a12 − a22a22], $E = \begin{bmatrix} a_{11} & -a_{21} \\ a_{12} & +a_{22} \end{bmatrix}$E = [a11 − a21a21a12 + a22a22]

Which of the matrices among $B, C, D,$ and EB, C, D, and E have the same determinant as that of the matrix AA, for any real numbers $a_{11}, a_{12}, a_{21}, a_{22}$ a11, a12, a21, a22?
[Hint: Try to observe whether $B, C, D,$ B, C, D, and EE can be obtained from AA by row or column operations of Type 22 mentioned in the key points.]

○
BB and DD
○
BB and EE
○
BB and CC
○
DD and EE
○
CC and EE

No, the answer is incorrect.
Score: 0
Feedback:
• B can be obtained by a row operation of Type 2.
• C can be obtained by column operation of Type 2.
• D cannot be obtained by a row operation or column operation of Type 2. Check the determinant of D and see whether it matches with determinant of A or not.
• E cannot be obtained by a row operation or column operation of Type 2. Check the determinant of E and see whether it matches with determinant of A or not.
Accepted Answers:
BB and CC

1 point

If all the elements of a $3 \times 33 \times 3$ real matrix AA are the same, then which of the following is (are) correct?

[Hint:
∵ If two rows or two columns of a matrix are equal, then the determinant of the matrix is 0.
∵ $det(A) = det(A^T)$det(A) = det(AT), where A^T AT denotes the transpose of AA].

☐ Determinant of matrix AA is 00.

☐ Determinant of matrix AA cannot be determined from the given information.

☐ Determinant of matrix AA will be the sum of the elements of a row.

☐ Determinant of matrix $A + A^T$ A + AT is 00, where A^T AT denotes the transpose of AA.

☐ Determinant of matrix $A + A^T$ A + AT cannot be determined from the given information, where A^T AT denotes the transpose of AA.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Determinant of matrix AA is 00.
Determinant of matrix $A + A^T$ A + AT is 00, where A^T AT denotes the transpose of AA.

1 point

Let $A, B,$ A, B, and CC be $3 \times 3$3 × 3 real matrices. Which of the following options is (are) correct?
[Hint: ∵ $det(A) = det(A^T)$det(A) = det(AT), where A^T AT denotes the transpose of AA.
∵ $det(AB) = det(BA)$det(AB) = det(BA).]

☐ $det(ABC) = det(A)det(B)det(C)$det(ABC) = det(A)det(B)det(C)

☐ $det(A^3) = det(A)^3$ det(A3) = det(A)3

☐ $det(A + B + C) = det(A) + det(B) + det(C)$det(A + B + C) = det(A) + det(B) + det(C)